

Department of VLSI Design and Systems Engineering
Digital Design MVPC 202 (1st Sessional) ; 25/09/2025

M.Marks :30

Time: 50 minutes.

Design a combinational circuit whose inputs are two 8-bit unsigned binary integers X and Y and a control signal MIN/MAX. The output of circuit is an 8-bit unsigned binary integer Z such that:

Q.I $Z = \min(X, Y)$ if MIN/MAX =1 and $Z = \max(X, Y)$ if MIN/MAX =0 [15]

Q.II Design logic circuit using full adders to obtain the following arithmetic functions on two 4-bit numbers A and B [15]

$$F = A$$

$$F = A + 1$$

$$F = A + B$$

$$F = A + B + 1$$

$$F = A - B - 1$$

$$F = A - B$$

$$F = A - 1$$

$$F = A$$